Reservoirs modify river thermal regime sensitivity to climate change: a case study in the southeastern United States

Yifan Cheng[1], Nathalie Voisin[2,1], John R. Yearsley[1], Bart Nijssen[1,*]

1. Department of Civil and Environmental Engineering, University of Washington, Seattle, WA 98195, USA

2. Pacific Northwest National Laboratory, Seattle, WA 98109, USA

[*]Corresponding author: Bart Nijssen (nijssen@uw.edu)

Key points:

- Summer stream temperatures in regulated rivers are lower than those in unregulated rivers, but they are more sensitive to climate change
- Reservoirs with long residence times will be more stratified in the future, but the effects on tailwater temperature will dissipate faster
- Simulated summer stream temperatures in regulated rivers are more sensitive to errors in hydrology than those in unregulated rivers

Abstract

Seasonal thermal stratification in reservoirs changes the thermal regime of regulated river systems as well as stream temperature responses to climate change. Cold releases from reservoir hypolimnion can depress downstream river temperature during warm seasons. Recent large-scale climate change studies on stream temperature have largely ignored reservoir thermal stratification. In this study, we used established models to develop a framework which considers water demand and reservoir regulation with thermal stratification, and applied this model framework to the southeastern US. About half of all 271 reservoirs in our study area retain strong thermal stratification by the 2080s (2070-2099) under RCP8.5 even as median residence times decrease to 60 days from 69 days in the historic period (1979-2010). Reservoir impacts on downstream temperatures become slightly weaker in the future because of higher air temperature and stronger solar radiation. We defined a "cooling potential" to quantify the thermal energy that a water body can absorb before exceeding a water temperature threshold. In the future, higher river temperatures will reduce the cooling potential for all river segments, but more so for river segments minimally impacted by thermal stratification. Reservoir impacts on cooling potential remain strong for river segments downstream of reservoirs with strong thermal stratification. We conducted a sensitivity analysis to evaluate the robustness of our findings to errors in the hydrological simulations. While river segments subject to reservoir regulation are more sensitive to errors in hydrology than those without regulation impacts, our overall findings do not materially change due to these errors.

Plain Language Summary

River temperature is influenced by both climate change and human activities, especially dam regulation. Large dams impound deep reservoirs where only the top layer is warmed while the bottom stays cold. This phenomenon, with colder, heavier water at the bottom and warmer, lighter water on top, is known as thermal stratification. Large reservoirs usually release water from the bottom and thus cool downstream river temperature in the summer. In this study, we use computer models to simulate river flow and temperature while considering the influence of reservoir regulation and thermal stratification. Thermal stratification will strengthen for about half of the 271 reservoirs in the southeastern US, but the cold outflow warms up faster due to global warming. Environmental agencies regulate maximum allowable river temperature, constraining a river's ability to provide cooling water for power plants. The cool release from the bottom layer provides more downstream cooling potential, i.e., the energy that a river can absorb before exceeding the maximum temperature, than would be possible in an unregulated river. In the future, the cooling potential will decrease as the climate warms, especially in unregulated rivers. While our models are subject to errors, these errors do not change our overall findings.

1. Introduction

The objective of the paper is to advance our understanding of the effects of climate change on thermal regimes in heavily regulated rivers and of climate change as a stress multiplier on currently existing water resources infrastructure. We are particularly interested in large spatial domains, as opposed to single reach or small basin studies, to capture the river thermal constraints on regional power networks as well as aquatic ecosystems. Here, we describe a study in the southeastern United States (SEUS), where the power sector is particularly vulnerable to projected changes in stream temperature (van Vliet et al., 2016; Liu et al., 2017). The study region coincides roughly with the service area of the SERC Reliability Corporation, which is responsible for the reliability and security of the electric grid across the southeastern and central regions of the United States. The study region contains a number of highly regulated river systems with more than 300 dams. One third of the region's electricity generation portfolio depends on thermoelectric plants with once-through cooling that require large amounts of water for supporting power operations (Averyt et al., 2013). Once-through cooling systems are legally-mandated to not produce effluent that exceeds certain environmental temperature thresholds. However, historical stream temperatures are already close to environmental thresholds as we will show in Section 3.2.

Previous studies have generally shown that projected increases in stream temperature under climate change will put great pressure on potential electricity generation globally (van Vliet et al., 2012, 2013, 2016) and over regions of the U.S. (Bartos and Chester, 2015; Boehlert et al., 2015; Liu et al., 2017; Miara et al., 2017; Zhang et al. 2020) by reducing the cooling efficiency of thermoelectric plants. Power plants face capacity derating and resulting power outages if their thermal effluent temperature exceeds specific environmental thresholds (Raptis et al., 2016;

McCall et al. 2016). This generation curtailment risk is particularly relevant for power plants using once-through cooling techniques because they release thermal effluent directly into nearby river systems. Liu et al. (2017) showed that the average generating capacity in the U.S. is expected to decline by up to 12% by 2060 if environmental regulations are enforced, with the power system in the eastern U.S. most vulnerable to increases in stream temperature (van Vliet et al., 2012, 2016; Liu et al., 2017).

Reservoir regulation can change a river's thermal response to surface meteorology and its thermal sensitivity to climate change. Manmade reservoirs can modify a river's thermal regime in two ways: 1) reservoirs store a large amount of water with a smaller surface area to volume ratio than a regular river reach and hence modify the thermal response to surface energy fluxes; and 2) deep reservoirs thermally stratify on a seasonal basis and store cold water for later release (Chapra, 1997). Multiple catchment-scale studies have investigated the impacts of simple reservoir systems on downstream river temperatures in a number of locations around the world, e.g., Canada (Maheu et al., 2016), China (Cai et al., 2018), Europe (Arora et al., 2018; Kędra & Wiejaczka, 2018), and the United States (Lowney, 2000). These studies show that seasonal thermal stratification has significant impacts on downstream river temperatures. Stratification results in a density gradient that inhibits mixing between the colder bottom layer (hypolimnion) and the warmer upper layer (epilimnion). Because many reservoir releases (e.g., for hydropower) are made from the bottom layer, summer stream temperatures are often cooler downstream of reservoirs.

While a number of large-scale stream temperature studies (Isaak et al., 2012; Mantua et al., 2010; van Vliet et al., 2012; Boehlert et al., 2015; Li et al., 2015; Strzepek et al., 2015; Sun et al., 2015, Zhang et al. 2020) accounted for the effects of reservoir regulation, most of them did not

explicitly consider seasonal thermal stratification and therefore underestimate the effects of regulation on stream temperature, particularly downstream of large reservoirs. For example, Li et al. (2015) showed a warm bias of over 10ºC in summer stream temperature downstream of Hoover Dam and Yearsley et al. (2019) showed a warm bias of as much as 8°C downstream of a reservoir in the Connecticut river basin. Those studies that did include the effects of stratification, including recent studies by Boehlert et al. (2015) and Strzepek et al (2015), did not explicitly evaluate reservoir impacts on river temperature further downstream.

Detailed models exist to simulate reservoir temperatures in regulated river systems, e.g., CE-QUAL-W2 (Cole and Wells, 2015), WQRRS (USACE-HEC, 1986) and HEC-5Q (Willey, 1986). These models have been used mainly in single-reservoir studies (Gelda et al., 1998; Hanna et al., 1999), in part because of their extensive data requirements. CE-QUAL-W2 (Hanna et al., 1999) and WEAP (Rheinheimer et al., 2014) have also been used to study temperature control device applications.

Water resources management requires coordinated decisions at the basin level and consequently needs tools and models that can account for the combined actions and effects of multiple reservoirs. Many regional applications, such as power system planning, require the ability to examine coordinated impacts across an even larger region, consisting of multiple river basins. Boehlert et al. (2015) and Niemeyer et al. (2018) developed simplified modules that represent reservoirs as two-layer systems that account for thermal stratification and incorporated these modules into distributed stream temperature models, that can represent the aggregate effect of tens or even hundreds of reservoirs.

In this study, we include the reservoir module of Niemeyer et al. (2018) in a regional model setup to examine climate change effects on stream temperature across a large region with

hundreds of reservoirs. We account explicitly for reservoir regulation, including the effects of seasonal thermal stratification, on system-wide stream temperatures, enabling us to evaluate the individual and joint contributions of climate change and reservoir impacts on river thermal regimes. We focus on thermal regimes during the summer when high air temperature, high water demand, and low streamflow coincide in our study region and when thermal stratification is strongest and has the greatest impact on downstream river temperature.

While a large-domain model setup allows for a system-wide evaluation of the thermal response and thermal sensitivity to climate change, large-domain model applications are typically subject to larger errors than small-domain or single-site applications. These larger errors result from process simplifications, limited information about site characteristics, and model parameters that cannot always be calibrated across all sites in a consistent manner, because of limited observations and computing resources. To address this general shortcoming of large-domain model implementations, we conclude this paper with a sensitivity experiment to assess the robustness of our findings to errors in the hydrological simulations.

2. Methods

Our process-based modeling approach uses a series of established models (Figure 1). It consists of a large-scale, spatially-distributed hydrological model (Variable Infiltration Capacity or VIC; Liang et al., 1994; Hamman et al., 2018), a river routing model (Model for Scale Adaptive River Transport or MOSART; Li et al., 2013), coupled to a spatially-distributed water management model (WM; Voisin et al., 2013a, 2017), and a stream temperature model (River Basin Model or RBM; Yearsley, 2009; 2012) that includes a two-layer reservoir thermal stratification module (2L; Niemeyer et al., 2018). We can disable the WM and 2L modules to simulate unregulated

147 river conditions (solid frames in Figure 1), which we also refer to as the unregulated model
148 setup. Historic meteorological forcings and downscaled climate change projections are pre-
149 processed using MetSim (Bennett et al., 2020) so that they can be used as input to our model
150 chain.

151 2.1. Study area

152 The study region (SEUS; 1.19 million km^2, Figure 2) was selected because it coincides roughly
153 with the service area of the SERC Reliability Corporation. We adjusted the area in some places
154 to capture river basin boundaries. The SEUS includes large river basins that cover all or parts of
155 Florida, Georgia, Alabama, Mississippi, Louisiana, Texas, Oklahoma, Arkansas, Missouri, Iowa,
156 Illinois, Kentucky, Tennessee, Virginia, North Carolina, and South Carolina. Over the past
157 century, more than 300 dams and locks were constructed in the area. Some of these are run-of-
158 river dams or navigation locks, which we excluded, because they do not have a significant effect
159 on streamflow or stream temperature. We explicitly simulate 271 major reservoirs based on
160 information in the Global Reservoir and Dam Database (GRanD; Lehner et al., 2011). Most of
161 the dams in the region were built in the headwaters rather than on the mainstem of the major
162 rivers, with the exception of the Tennessee River, which has seven mainstem reservoirs (Table
163 S1, with 'S' indicating material in the Supporting Information).

164 2.2. Meteorological forcings

165 Gridded meteorological forcings were used as input to our model chain. We used gridMET
166 (Abatzoglou, 2013) for the historical period (1979 – 2010). For the future climatological period
167 (2070 – 2099, hereafter referred to as the 2080s), we used projections from the Coupled Model
168 Intercomparison Project, Phase 5 (CMIP5; Taylor et al., 2011), downscaled using the

Multivariate Constructed Analog (MACA; Abatzoglou and Brown, 2012) method. Since gridMET is the training dataset for MACA, consistency between the historical and future projections is ensured. The gridMET and MACA datasets were regridded from 1/24º to 1/8º latitude-longitude resolution for our application. We used an ensemble of 20 global climate models (GCMs; Table S2) and one representative concentration pathway (RCP; RCP8.5) for a total of 20 future scenarios. Daily gridMET and MACA data, i.e., daily maximum and minimum air temperature, precipitation and wind speed, were disaggregated to the 3-hourly VIC model time step using MetSim v2.0.0 (Bennett et al., 2020). Other subdaily meteorological forcings, including relative humidity, surface air pressure, and incoming shortwave and longwave radiation were estimated by MetSim using algorithms from the Mountain Micro Climate Simulator (MTCLIM) (Bohn et al., 2013; Thornton & Running, 1999). Wind speed was assumed constant throughout the day.

2.3. Streamflow and stream temperature

2.3.1. Hydrology and runoff

We used VIC version 5 (VIC5) (Hamman et al., 2018) at a 1/8º spatial resolution to simulate the hydrology of the SEUS. Model parameters were taken from Maurer et al. (2002) and modified over part of the domain to reduce a high streamflow bias in the South Atlantic Region. VIC was run at the 3-hourly time step and fast and slow response runoff were output at the daily time step to be consistent with the temporal resolution of the river routing and stream temperature models. We initialized the VIC simulation using a 32-year spin-up run forced by historical meteorological data to allow sufficient time for model states, e.g., deep layer soil moisture, to equilibrate.

2.3.2. Streamflow and reservoir regulation

MOSART-WM (Voisin et al., 2013a) is a fully-coupled spatially-distributed river routing and water management model that accounts for reservoir operations and multi-sectoral water demands to simulate regulated streamflow over large regions (Voisin et al., 2013a, 2017; Hejazi et al., 2015; Zhou et al., 2018). We implemented MOSART (Li et al. 2013) with and without the water management module (WM; Voisin et al., 2013a, 2017) at a daily time step with a 1/8° spatial resolution. WM estimates daily reservoir releases based on monthly storage targets and accounts for daily constraints such as environmental flow, minimum and maximum storage, and water demands. We substituted WM's original computation of dynamic storage targets at the end of each month (Voisin et al., 2013a) with specified, observed monthly storage targets calculated from storage or reservoir height observations or based on a guide curve (see Text S1 for details, including Figures S1, S2, S3, S4, and Tables S3, S4, S5, and S6). We collected guide curves or calculated them from observed storages or reservoir elevations for 92 reservoirs in the study area based on information from the Tennessee Valley Authority (TVA), United States Geological Survey (USGS), and U.S. Army Corps of Engineers (USACE) (data source for each reservoir is summarized in Table S5). Storage targets for reservoirs without observations were estimated based on their primary regulation objective and the relative degree to which they were filled on average (Text S1). In addition to the storage targets, inputs for MOSART-WM included runoff from VIC, reservoir information from the GRanD Database, and 2010 level water demand as used in Voisin et al. (2017) and Zhou et al. (2018). We used fixed water demand from year 2010 for all years to isolate the effects of climate change from other anthropogenic influences which can significantly impact the hydrologic response (Voisin et al. 2013b, 2016).

2.3.3. Stream temperature

RBM is a one-dimensional stream temperature model which solves the energy balance equation for surface energy fluxes in the river channel (Yearsley, 2009; 2012). To account for seasonal thermal stratification, we used a module (2L) developed by Niemeyer et al. (2018) that simulates the temperature in two thermally distinct layers: an epilimnion (top layer) and a hypolimnion (bottom layer). This module uses inflow, outflow, and storage data for each reservoir as simulated by MOSART-WM. To calculate surface energy exchange between the water and the overlying atmosphere, RBM and RBM-2L (i.e. without and with reservoir representation) use output from VIC and MetSim. Headwater temperatures, i.e., the most upstream river temperature, which form a boundary condition when solving the energy balance equation for river systems, were estimated using a non-linear relationship with air temperature (Mohseni, 1998). Fitted parameters for the Mohseni equations were taken from Niemeyer et al. (2018). We assumed that all reservoir releases were made from the hypolimnion so our result represents a lower bound of possible river temperatures under reservoir regulations. RBM and RBM-2L were initialized separately by setting all model elements to the same water temperature and running the model for a one-year spin-up period.

2.4. Evaluation metrics

We examine the impacts of climate change on water availability, residence time, regulated stream temperature, and "cooling potential", a metric used to evaluate the compound impacts of water availability and stream temperature. Reservoir residence time was calculated as the mean annual storage divided by the mean annual outflow from that reservoir. In summer, reduced streamflow and increased stream temperature can both constrain the operations of thermoelectric power plants. The cooling potential (E^{cp} [W]) combines the effects of stream temperature and streamflow and represents the additional amount of energy that a body of water can absorb, e.g.,

from thermoelectric power plants, before exceeding a water temperature threshold. We define the cooling potential as

$$E^{cp} = \begin{cases} \rho C_p (T_{threshold} - T_{summer}) Q_{summer} & if\ T_{summer} < T_{threshold} \\ 0 & if\ T_{summer} \geq T_{threshold} \end{cases}, \quad (1)$$

where ρ is water density [kg/m^3], C_p is the heat capacity for water [4186 J/kg·°C], $T_{threshold}$ is a stream temperature threshold [°C], T_{summer} is the mean summer stream temperature [°C], and Q_{summer} is the mean summer streamflow [m^3/s]. For $T_{threshold}$ we used one single value for the entire region for a spatially consistent analysis. We set $T_{threshold}$ to 31°C, which is the median of the maximum stream temperature standards of all states in our study region as imposed by the United States Environmental Protection Agency (EPA). In other words, the cooling potential represents a river system's ability to absorb waste thermal energy from thermoelectric power plants if environmental regulations are enforced. We examine climate change impacts on the cooling potential not only at the sites of existing thermoelectric power plants, but for the entire regulated river system.

One of our goals is to quantify the effect of reservoir regulation on the thermal sensitivity of the system to climate change. To this end, we define a rate of change (ΔR) as

$$\Delta R = \left(\frac{|\Delta E^{cp}{}_{(reg,2080s-nat,2080s)}|}{|\Delta E^{cp}{}_{(reg,hist-nat,hist)})|} - 1 \right) \times 100\%, \quad (2)$$

where $\Delta E^{cp}{}_{(reg,t-nat,t)}$ is the change in cooling potential due to regulation during period t. The metric ΔR ranges from -100% to positive infinity. Positive values indicate that future climate will enhance the ability of regulation to maintain the cooling potential, while negative values for ΔR indicate that future climate will reduce the ability of regulation to maintain the cooling potential. In other words, negative ΔR means that rivers are less able to absorb waste heat before

exceeding the threshold temperature, potentially affecting cooling for thermal power plants and hence power generation under climate change.

2.5. Sensitivity Analysis

Large-domain model simulations are typically subject to larger errors than small-domain or single-site applications and these errors are often not easy to reduce. In our model chain, errors in regulated streamflow mainly result from biased runoff from the hydrological model and fixed monthly storage targets. To determine the robustness of our findings to errors in the hydrological simulations, we performed a sensitivity analysis.

As part of our climate change experiment we already created 20 individual hydrological simulations, each corresponding to the downscaled meteorological forcings of a single GCM, and each resulting in a different, but internally consistent streamflow. In the sensitivity analysis, we used these 20 different hydrological simulations as alternative streamflow scenarios, which we then combined with the meteorological forcings from a single GCM as input to the stream temperature model. The subsequent spread in river temperatures resulted directly from the spread in hydrologic conditions. We repeated this process for three different climate change scenarios selected from our 20 GCM simulations (high - HadGEM2-CC365, medium - CSIRO-Mk3-6-0, low - inmcm4; based on mean projected air temperatures) resulting in 60 (3×20) stream temperature simulations. The three temperature scenarios account for potential differences in sensitivity to hydrological error for different changes in air temperature. We ran this experiment using both regulated and unregulated model setups, resulting in a total of 120 (2×60) stream temperature simulations.

We designed a metric (ϕ) which relates the spread in stream temperature or cooling potential to the spread in hydrologic conditions. Large ϕ values indicate that changes in hydrological

conditions result in a large spread in simulated stream temperature or cooling potential. For stream temperature we calculated this metric as the ratio of the standard deviation (σ) of the projected increase in mean summer river temperature (ΔT), in unit of °C, to the coefficient of variation (CV) of mean annual streamflow (Q). The resulting ϕ-values for stream temperature are in unit of °C. We categorized river segments based on river size and whether they are subjected to reservoir regulation. River size is based on the historical mean annual regulated streamflow (Q) with each segment receiving a classification (l) of small ($Q \in$[0,50] m^3/s; l=1), medium ($Q \in$[50,100] m^3/s; l=2), or large ($Q \geq$100 m3/s; l=3). All river segments located downstream of reservoirs are subjected to reservoir regulations (m=true) while the rest are not (m=false). We calculated the metric for each group as follows

$$\phi_{i,l,m,r} = \frac{\sigma_{\widetilde{\Delta T}_{i,[\![j]\!],[\![k]\!]_{l,m},r}}}{CV_{\widetilde{Q}_{i,[\![j]\!],[\![k]\!]_{l,m},r}}} \quad (3)$$

$$\widetilde{\Delta T}_{i,j,k,r} = \Delta T_{i,j,k,r} - \overline{[\Delta T_{\iota,k,r}]_J} \quad (4)$$

$$\tilde{Q}_{i,j,k,r} = \frac{Q_{i,j,k,r}}{\overline{[Q_{\iota,k,r}]_J}} \quad (5)$$

where i denotes the climate change scenarios (i=high, medium, low), j denotes the streamflow scenario based on each GCM (j=1, 2, …, 20), the double brackets $[\![j]\!]$ indicate that we calculate the statistic across all values of j, k denotes the grid cells in our model domain, $[\![k]\!]_{l,m}$ denotes all grid cells within river size l and subjected to regulation scenario m, and r denotes the regulated and unregulated model setups, i.e., WM and 2L are enabled (r=reg) and disabled (r=unreg), respectively . In Equation 4 we removed $\overline{[\Delta T_{\iota,k,r}]_J}$, i.e., the mean value across 20 GCMs, from the temperature change, $\Delta T_{i,j,k,r}$ so that the hydrology-induced spread ($\widetilde{\Delta T}$) can be aggregated across all grid cells within each group. Similarly, we divided the $Q_{i,j,k,r}$ values by $\overline{[Q_{\iota,k,r}]_J}$, i.e.,

the mean across 20 GCMs (equation 5). $\phi_{i,l,m,r}$ quantifies the temperature spread caused by hydrologic errors, with larger $\phi_{i,l,m,r}$ values indicating that temperatures are more sensitive to errors in hydrology.

We related the calculated sensitivities to the errors in our hydrological simulations during the historic period. That is, we used $\phi_{i,l,m,r}$ and the hydrologic errors from model evaluation to estimate the resulting error in the stream temperature simulations as follows

$$\hat{\sigma}_{\widetilde{\Delta T}_{i,l,m,r}} = \phi_{i,l,m,r} \times \widehat{CV}_{\tilde{\mathrm{Q}}_{[\![\hat{k}]\!]_{l,m}}} \quad (6)$$

$$\tilde{Q}_{\hat{k}} = \frac{Q_{sim,\hat{k}}}{Q_{obs,\hat{k}}} \quad (7)$$

where $\hat{\sigma}_{\widetilde{\Delta T}}$ denotes errors in the stream temperature change resulting from errors in our hydrological simulations, $\hat{k}$ denotes corresponding grid cells with USGS observations ($\hat{k}$=1,2,…,111; $\widehat{[\![k]\!]}$=[1,2,…,111]), $Q_{sim,\hat{k}}$ and $Q_{obs,\hat{k}}$ denote simulated and observed mean annual streamflow for site $\hat{k}$, and the accent hat indicates estimates based on comparison with observations. Unlike the sensitivity analysis above with 20 hydrologic scenarios, we only have one historical scenario, so we calculated a representative value of hydrological errors for each group using all corresponding USGS sites. We divided the $Q_{sim,\hat{k}}$ values by $Q_{obs,\hat{k}}$ (equation 7) to aggregate multiple USGS sites within the same group.

We repeated the same analysis for cooling potentials (E^{cp}). Instead of $\widetilde{\Delta T}_{i,j,k,r}$ (equation 4), we used the projected relative changes of cooling potentials as follows:

$$\Delta r{E^{cp}}_{i,j,k,r} = \frac{[{E^{cp}}_{i,j,k,r}]_{2080s} - [{E^{cp}}_{k}]_{hist}}{[{E^{cp}}_{k}]_{hist}} \quad (8)$$

$$\widetilde{\Delta r{E^{cp}}}_{i,j,k,r} = \Delta r{E^{cp}}_{i,j,k,r} - \overline{[\Delta r{E^{cp}}_{\iota,k,r}]}_{j} \quad (9)$$

where subscript "hist" denotes historical periods with other subscripts as above, and ΔrE^{cp} represents the projected relative changes in cooling potential. Furthermore, we removed the mean of ΔrE^{cp} across 20 GCMs, similar to what we did for the mean summer temperature.

3. Results

3.1. Model evaluation

3.1.1. Streamflow

To evaluate the simulated streamflow and stream temperature in a regulated river system, we used USGS observations at 111 sites with both streamflow and stream temperature data (USGS, 2019; summarized in Table S7. Site selection was subject to two criteria: (1) each site had to have at least one-year of observations that overlapped with our historical simulation period and (2) the contributing area of the site had to be larger than the size of a single model grid cell (approx. 150 km^2). Sixty-three of the 111 sites were located downstream of reservoirs and therefore subject to reservoir regulation. The remaining 48 sites were only minimally affected by regulation and were used to evaluate whether poor performance at regulated sites was the result of poor performance of the hydrology model or the water management model. We used the relative bias at the annual time step and the Nash-Sutcliffe (NS) coefficient at the monthly time step to evaluate overall water availability and streamflow seasonality, respectively (Figure S5). Across the 111 sites with observations, the median relative bias in mean annual simulated streamflow was 0.05, which includes the effects of regulation for sites downstream of reservoirs. Sixty-three sites, representing a mixture of regulated and unregulated sites, had a relative bias with an absolute value less than 0.2. Seven had a relative bias with an absolute value greater than 0.5. The NS coefficient for the monthly flows shows lower values for sites with a larger relative

bias. In addition, the simulated flows generally show higher NS coefficients for sites that are not subject to regulation. The median NS coefficient for the 48 unregulated sites was 0.70 compared to 0.55 for the 63 regulated sites. Three unregulated sites and 17 regulated sites had an NS coefficient less than 0. The South Atlantic region generally had the worst performance as measured by the NS coefficient, in part because multiple sites (n=6) downstream of Buford reservoir on the Chattahoochee river showed relatively poor performance. The impacts of hydrologic errors on stream temperature simulations are further investigated through the sensitivity analysis in Section 4.3.

3.1.2. Reservoir storage

For the 92 reservoirs with guide curves (Section 2.3.2), we compared simulated storage to the specified guide curves. Because we specify the guide curves in MOSART-WM, the model storage is expected to closely follow these curves, unless streamflow is too low to fill the reservoir or daily release-constraints force a deviation from the specified guide curve. We calculated the normalized root mean square error (nRMSE) of multi-year mean monthly storage (12 mean monthly values based on the 32-year daily time series) and the relative bias of mean annual storage to evaluate seasonality and bias, respectively, in the thermal mass of the reservoirs. Simulated storage showed only a small relative bias and a small nRMSE for the 92 sites with guide curves, with median values of -0.01 and 0.06 respectively, indicating that the water balance model captures the volume and seasonality of the storage in the study region.

3.1.3 Stream temperature

In the SEUS, low flow occurs during the summer when it coincides with high air temperatures, resulting in high stream temperatures. Summer is also the time when reservoir stratification is

most pronounced. We therefore focus our evaluation on the mean summer river temperature (Figure S5), with summer defined as the three-month period from June to August.

Simulated mean summer stream temperatures generally show a cold bias with a median value of -0.7 ºC across the 111 sites with stream temperature observations. The mean bias was less than ±2 ºC at 87 of these sites. The bias was generally smaller at the unregulated sites (median bias -0.3 ºC, n=48) than at the regulated sites (median bias -0.9 ºC, n=63). For the 63 regulated sites, 45 had a bias less than 2 ºC. Nine regulated sites had a cold bias larger than 4 °C.

The Supporting Information contains site-specific information of model performance, including time series plots of observed and simulated streamflow (Figure S6) and stream temperature (Figure S7) for sites downstream of the five largest reservoirs with downstream temperature observations and a discussion of the source of hydrological errors (Text S2).

3.2. Historical analysis: Reservoir impacts typically maintain or reduce mean summer stream temperature

Figures 3a and 3b show, respectively, the simulated mean summer temperature for the historic period for the regulated river system and the impact of regulation on stream temperature.

Seasonal thermal stratification results in lower stream temperatures immediately downstream of large reservoirs and gradually attenuates further downstream. Smaller reservoirs, which do not stratify, have little to no impact on downstream river temperature.

Larger reservoirs with longer residence times tend to have stronger thermal stratification and impact downstream temperature more strongly. For example, summer outflow from Buford Reservoir (residence time of 402 days) on the Chattahoochee River is 15.2 °C cooler in the regulated than in the unregulated scenario. Youghiogheny Reservoir on the Youghiogheny River has a shorter residence time (208 days) and a smaller cooling effect of 7.4 °C. Reservoirs with

even shorter residence times often have little to no impact on mean summer stream temperature as shown by a number of reservoirs on the Catawba River.

Reservoirs with larger outflows depress stream temperature for a longer distance downstream of the dam because their greater thermal mass makes them (1) less sensitive to surface energy fluxes and (2) less affected by mixing of tributary flows. For example, Deep Creek Reservoir on the Youghiogheny River (upstream of Youghiogheny Reservoir) has a small mean summer outflow of 3 m^3/s and a tailwater temperature that is 12 ºC colder in the regulated case. This temperature difference reduces to only 3 ºC over a distance of less than 18 km. In contrast, J. Strom Thurmond Reservoir on the Savannah River has a much larger mean summer outflow of 111 m^3/s and the temperature difference due to regulation decreases from 11 °C to 4 °C over about 100 km. So, while residence time affects seasonal thermal stratification and therefore the tailwater temperature, the outflow volume affects how far downstream this temperature depression persists.

3.3. Climate change impacts on regulated river temperature

In general, stream temperature in the southeast is projected to increase, mainly resulting from increased air temperature (Figure 4a). Stream temperature increases are larger in the northern part of the study domain and smaller in the southern coastal region. The spatial pattern of stream temperature increase is strongly correlated with the spatial pattern of air temperature increase (Figure S9). Even though reservoir residence time decreases by the 2080s, reservoir regulation still results in colder stream temperatures downstream of reservoirs with seasonal thermal stratification than would occur in the absence of these reservoirs (Figure 4b). However, the cooling effect of reservoirs on downstream river segments slightly decreases by the 2080s. Figure 4c shows the change in the median temperature effect of reservoirs between the 2080s

and the historic period. Red colors indicate less cooling due to regulation in the future than in the past. Blue indicates greater cooling. By the 2080s, reservoir impacts on downstream river temperatures attenuate faster in the downstream direction, i.e., tailwater temperatures return to unregulated temperatures in a shorter distance. Even if a reservoir releases cool water from the hypolimnion, this cooling effect on tailwater temperature will dissipate faster under climate change as a result of greater surface energy fluxes. In other words, under climate change summer stream temperatures in regulated rivers increase faster than in unregulated rivers and are more sensitive to climate change.

3.4. Climate change modulates reservoir impacts through water availability and residence time

We evaluated the projected changes in reservoir outflow and residence time by the 2080s, to separate the contributions of changes in water availability from changes in thermal dynamics on the overall changes in stream temperature. Because we maintained the same storage targets in the future as during the historical period, the median projected changes for reservoir storage are within 5% for all reservoirs.

Residence times are projected to decrease for almost all reservoirs. Because we imposed the same storage targets in the future, projected increases in reservoir outflow lead to shorter residence times (Figure S10). Projected increases in reservoir outflows are generally greater in the north and coastal regions than in the central and southwestern parts of the study domain. The 20 GCM projections show strong agreement in the sign of the change signal. For most reservoirs more than 16 out of 20 models agree that outflows will increase, and residence times will decrease (Figure S11). This increase in outflow mostly results from increased precipitation in the

region. Although evapotranspiration also increases, it does not compensate for the increase in precipitation.

3.5. Cooling potential will decrease under climate change

Cooling potential (Equation 1) is influenced by both streamflow and stream temperature. During the historical period, cooling potential is greater in the regulated scenario for river segments downstream of reservoirs with strong thermal stratification, e.g., the Savannah River and the upper reaches of the Tennessee River (green in Figure 5d). Downstream of reservoirs with little to no thermal stratification, the cooling potential tends to be lower in the regulated scenario because reservoir operations and water withdrawals lead to lower summer streamflow than in the unregulated scenario, e.g., mainstem Ohio River (blue in Figure 5d).

Stream temperature is projected to increase in all GCM projections, contributing to a reduction in cooling potential. At the same time, streamflow is projected to increase in most GCMs, increasing cooling potential by increasing the thermal mass. The combined effects result in a net decrease in projected cooling potential during the summer by the 2080s for most GCM projections under RCP 8.5 (blue in Figure 5c). Therefore, the change in cooling potential is dominated by projected increases in stream temperature.

Larger rivers tend to show a greater loss in cooling potential than smaller rivers because of their greater thermal mass, e.g., the Ohio River, the largest river in our study region. Over most of its length, the Ohio River will lose over 10 GW (10^4 MW) of cooling potential by the 2080s under RCP8.5. Regulation impacts on cooling potential for the historical period and the 2080s under RCP8.5 are shown in Figure 5d and 5e, respectively. Rivers with reservoirs that experience strong thermal stratification (Figures 3b and 4b) have more cooling potential under the regulated scenario in both the historical and future periods. Rivers with reservoirs that do not stratify, have

457 less cooling potential in the regulated scenario (Figure 5d and 5e). For these rivers, regulation
458 reduces cooling potential because of water withdrawals and lower summer streamflow.
459 Figure 5f summarizes whether the impact of regulation on cooling potential will increase or
460 decrease in the future. If ΔR value is close to 0%, regulation impacts on cooling potential will
461 persist under climate change. If ΔR approaches -100%, regulation impacts will mostly disappear.
462 For river segments downstream of reservoirs with strong seasonal thermal stratification, the
463 regulation impact on cooling potential remains strong in the future, with ΔR mostly larger than -
464 25% (Figure 5f). For river segments minimally impacted by upstream thermal stratification,
465 regulation impacts on cooling potential decrease dramatically by the 2080s under RCP8.5 ($\Delta R <$
466 -75%; Figure 5f). For these river segments, the effect of regulation on cooling potential largely
467 disappears by the 2080s.

468 # 4. Discussion

469 ## 4.1. Reservoir residence time and thermal stratification

470 We used the mean summer temperature difference between the epilimnion and the hypolimnion
471 (ΔT_{e-h}) as a measure of thermal stratification, with larger values indicating stronger
472 stratification. To quantify the impacts of reservoir residence time on thermal stratification, we
473 compared simulated residence times and corresponding ΔT_{e-h} values for all 271 (Figure 6a).
474 Thermal stratification is weak when the reservoir residence time is shorter than 20 days. For
475 reservoirs with a simulated residence time greater than 20 days, we fitted a log-linear regression
476 model for ΔT_{e-h} as a function of residence time. Confidence intervals (95%) for the predicted
477 ΔT_{e-h} values were calculated based on Devore (2011) and are shown as dashed lines in Figure 6.
478 For the three reservoirs for which we have observations of reservoir storage, outflow and

reservoir temperatures with depth, we calculated the mean residence time and ΔT_{e-h} values (stars in Figure 6a). Of these three, only Cherokee Reservoir had a residence time longer than 20 days. Fort Loundon and Guntersville reservoirs have short residence times of 8 and 12 days, respectively, and their thermal stratification is weak.

By the 2080s under RCP8.5, thermal stratification will be stronger in about half the reservoirs. This finding is somewhat counterintuitive. Because we assume that the future guide curves will remain the same as for the historic period, increased streamflow results in reduced reservoir residence time by the 2080s (arrows generally point left in Figure 6b) and should therefore lead to weaker stratification (Figure 6a). Median residence time of all 271 reservoirs is projected to decrease from 69 days (historic) to 60 days (2080s). This decrease in residence time should be accompanied by a median decrease in thermal stratification of 0.7ºC (Figure 6a). While some smaller reservoirs with shorter residence times show the expected decrease in thermal stratification, median thermal stratification for all reservoirs is almost identical in the historic and future scenarios and about half of the reservoirs (135 out of 271) are more stratified by the 2080s, especially larger reservoirs with longer residence times (black arrows to the right in Figure 6b).

This counterintuitive result occurs because thermal stratification is also influenced by surface energy exchange during the summer season. In general, thermal stratification increases with increasing air temperature for reservoirs with similar residence times (Figure 6a). As air temperature increases along with stronger solar radiation, the epilimnion temperature increases faster than the hypolimnion temperature because the latter is warmed only by advective and diffusive energy exchange with the epilimnion. As a result, seasonal thermal stratification in our study region is stronger as air temperature increases, resulting in a change in slope in the

relationship between residence time and thermal stratification under climate change (contrasting red and blue lines in Figure 6b).

4.2. Stream temperature and cooling potentials downstream of reservoirs

Cold hypolimnetic releases equilibrate with the environment as a result of surface heat fluxes. Reservoir impacts on river thermal regimes dissipate completely when stream temperatures reach the unregulated river temperature. The distance it takes from the reservoir outlet to reach the unregulated river temperature is influenced by both streamflow and surface energy fluxes.

We used the model simulations performed in support of the sensitivity analysis (Section 2.5) to evaluate the relative contribution of streamflow and surface meteorology to the tailwater thermal regimes. In reality, it is not feasible to directly calculate the distance to reach the unregulated river temperature, because reservoir outflow might not reach this temperature before it flows into a downstream reservoir or merges with another stream. Instead, we evaluated the impacts of surface meteorology by comparing downstream temperatures at a fixed distance downstream of a reservoir for the cold, medium and hot scenarios (Section 2.5).

Surface meteorology has a strong impact on tailwater thermal regimes. For example, median outflow temperature increases 6.8°C, 7.4°C and 9.0°C under cold, medium, and hot scenarios, respectively, 50 km downstream of Youghiogheny Reservoir (Figure 7). Even though the thermal stratification is stronger in the hot scenario, the temperature of the released water also increases more rapidly downstream of the reservoir. Regulated stream temperature is relatively insensitive to changes in streamflow, in part because we use the same guide curves for all scenarios. In spite of the large spread in simulated streamflow, the spread in outflow temperature from Youghiogheny Reservoir across the twenty different hydrological inputs reduces from 3°C to about 0.2°C within 50 kilometers (Figure 7).

Immediately downstream of reservoirs, cooling potential is impacted by changes in stream temperature as well as changes in streamflow. Compared to unregulated rivers, cold outflow from stratified reservoirs increases the cooling potential (Eq. 1), and the amount of flow (Q) is also different than in unregulated rivers. Once the stream temperature approaches the unregulated river temperature, regulation impacts cooling potential only by altering streamflow.

4.3. Sensitivity to errors in hydrology

In this section, we limit our analysis to river segments subject to regulation (*m*=true) and compare their behavior in the regulated and unregulated model setups (*r*=reg, unreg). Analysis (not shown) revealed that the sensitivity of river segments not subject to regulation (*m*=false) is similar to the sensitivity of river segments subject to regulation in the unregulated model setup (*m*=true, *r*=unreg).

Figure 8a shows the projected increases in mean summer stream temperature (ΔT) as well as our estimate of the error ($\hat{\sigma}_{\widetilde{\Delta T}}$). This error only reflects the effects of errors in the hydrological simulations. Each panel shows $\Delta T \pm \hat{\sigma}_{\widetilde{\Delta T}}$ (solid line) and $\Delta T \pm 2\hat{\sigma}_{\widetilde{\Delta T}}$ (dashed line) for the regulated and unregulated scenario. Projected increases in mean summer temperature range from +2°C to +6°C while the hydrology-induced errors ranges from 0.09°C to 1.33°C. The hydrology-induced error in stream temperature is greatest immediately downstream of reservoirs (Figure 7) and gradually decreases as the river flows downstream.

River segments in the regulated model setup (*m*=true, *r*=reg) are more sensitive to errors in hydrology ($\hat{\sigma}_{\widetilde{\Delta T}}$ ranges from 0.27°C to 1.33°C) than the same segments in the unregulated model setup (*m*=true, *r*=unreg; $\hat{\sigma}_{\widetilde{\Delta T}}$ ranges from 0.09°C to 0.11°C). This is especially true for small river segments which have greater ϕ values than larger rivers (*m*=true, *r*=reg). While ϕ values

547 differ across river size and whether we represent the effects of regulation (*r*=reg, unreg), they are
548 similar across all three climate change scenarios.

549 Figure 8b shows the same information as Figure 8a, but for the change in cooling potential rather
550 than change in mean summer temperatures. Errors in the hydrological simulations have the
551 largest effect on the relative change in cooling potential for small rivers ($\bar{Q}$<50m^3/s) under the
552 cold scenario. In that case, mean projected relative changes in cooling potential ($\overline{\Delta rE^{cp}}$)
553 are -21.0% and -31.0% while the hydrology-induced errors ($\hat{\sigma}_{\widehat{\Delta rE^{cp}}}$) are 46.0% and 57.6% for
554 regulated and unregulated model setups, respectively (Figure 8b). In all other cases, the $\overline{\Delta rE^{cp}}$
555 values are larger than the $\hat{\sigma}_{\widehat{\Delta rE^{cp}}}$ (upper bound of solid line is below zero in Figure 8b), implying
556 that the sign of the change is not sensitive to the errors in hydrology in our simulations.
557 Uncertainties in the relative change in cooling potentials decrease with increasing river size and
558 increases with projected air temperature. Reservoir regulation does not have a strong impact on
559 the sensitivity of the relative change in cooling potentials to hydrologic errors, with similar
560 $\phi_{\widehat{\Delta rE^{cp}}}$ values for regulated and unregulated river segments (Figure 8b).
561 Overall, we conclude that the errors in our hydrological simulations affect our quantitative
562 estimates of changes in stream temperature and cooling potential in the following way. Errors in
563 our change estimates are greater in the regulated model setup (*r*=reg) than the unregulated setup
564 (*r*=unreg) and are largest for the smaller river segments. However, in general, the errors do not
565 affect the sign of the change signals nor the relative ranking of the change signals across river
566 sizes, regulation status or climate scenario. The effect of hydrological simulations on changes in
567 mean summer stream temperature for river segments not affected by regulation (*m*=false) is
568 small.

5. Conclusions

River temperatures play an important role in aquatic ecosystems and affect the efficiency of once-through thermoelectric power plants. In this study, we applied a physically-based model chain to simulate hydrology and stream temperature for a large part of the southeastern United States. Seasonal thermal stratification was explicitly represented. We used an ensemble of future climate forcings to quantify the impacts of climate change on the summer stream temperatures and cooling potential of southeastern rivers for the 2080s. The model chain was applied with and without water management to quantify changes in the effect of regulation on cooling potential as a result of climate change. We evaluated reservoir flow, storage, and stream temperature, along with their interactions and the overall response to climate change, which has not been done before. The novel cooling potential approach provides a new opportunity to support multi-sectoral long-term planning. We conducted a sensitivity analysis to quantify the robustness of our findings to errors in the hydrological simulations. Our approach has the advantage that it provides a consistent evaluation over a region that spans multiple watersheds, where water management decisions may affect stream temperatures. This region may be defined by institutional governance for natural conservation, electricity operations, and other activities. The approach is applicable to other, individual watersheds, although we would recommend using operational water management models for the basin of interest to simulate the hydrology.

We would like to highlight a few areas for further research and improvement. Reservoir operations are more complex than the generic operating rules or rule curves used in this study. More complex rules that use foresight and optimization across the system could be considered, especially if these rules are allowed to change with time. An explicit link with power system models could further our understanding of the response to local stresses as well as feedbacks

592 onto electricity operations and downstream river systems. For example, Miara et al. (2018)
593 demonstrated the dynamic link with power system operations but lacked the representation of
594 reservoir operations and their impact on stream temperature. Finally, to isolate the effects of
595 changes in meteorological forcings in a changing climate, we kept land use and urbanization
596 constant in our model simulations, even though they may impact both hydrology and stream
597 temperature. Explicit representation of these changes may enhance our understanding of the co-
598 evolution of complex interconnected systems.
599 Major findings are as follows:
600 • Reservoir regulations, and the resulting seasonal thermal stratification, influence
601 downstream river temperatures. Reservoirs with longer residence times are more
602 stratified and store cold water from earlier seasons in their hypolimnion. They therefore
603 have stronger impacts on downstream river temperatures. Among reservoirs with similar
604 residence times, thermal stratification tends to be stronger in warmer locations where
605 reservoirs are subject to stronger surface energy fluxes.
606 • Reservoir regulation changes how river temperature responds to climate change,
607 especially through changing thermal stratification. Summer stream temperatures in
608 regulated rivers are lower than those in unregulated rivers, but they are more sensitive to
609 climate change. By the 2080s under RCP8.5, compounded impacts of higher air
610 temperature and shorter residence times result in stronger thermal stratification for over
611 half of all reservoirs in the SEUS, especially for reservoirs with longer residence times.
612 Thermal stratification impacts on downstream river temperatures mostly persist but are
613 slightly weakened, because of higher air temperature and stronger surface energy inputs.

- Cooling potential is the energy required to warm rivers to a threshold temperature and represents a river system's ability to absorb waste thermal energy from thermoelectric power plants if environmental regulations are enforced, indicating the compound impacts of both streamflow and stream temperature. For rivers experiencing strong thermal stratification impacts, regulation increases cooling potential immediately downstream of reservoirs because the cooling impact on downstream river temperature dominates the signal. For rivers minimally impacted by thermal stratification, regulation decreases cooling potential because of lower summer streamflow under reservoir regulation.
- Cooling potential is projected to decrease for all river segments under climate change. The Ohio River will lose over 10 GW cooling potential for most of its length by the 2080s under RCP8.5. Regulation changes how cooling potential responds to climate change. Regulation impacts on cooling potential remain strong when rivers are strongly influenced by thermal stratification ($\Delta R > -25\%$, Figure 6f) because cooling impacts on downstream river temperatures mostly persist under climate change. For rivers minimally impacted by thermal stratification, the magnitude of regulation impacts on cooling potential decreases dramatically ($\Delta R < -75\%$, Figure 6f) because river temperatures are higher by the 2080s under RCP8.5.
- Sensitivity analyses show that our findings about climate change effects on mean summer river temperature and cooling potentials are relatively insensitive to errors in the hydrological simulation. Mean summer temperatures in smaller rivers tend to be more sensitive to errors in hydrology than those in larger rivers. Mean summer temperatures in regulated river segments are much more sensitive to hydrological errors than those in

unregulated river segments. However, regulation has little impact on the sensitivity of changes in cooling potential to hydrological errors.

Acknowledgement

This project was funded in part by NOAA grant NA14OAR4310250 and NSF grant EFRI-1440852 to the University of Washington. We also wish to thank the Tennessee Valley Authority for providing data, and Xiao Zhang at Pacific Northwest National Laboratory for help with routing and water management model setup. We acknowledge the World Climate Research Programme's Working Group on Coupled Modelling, which is responsible for CMIP, and we thank the climate modeling groups (listed in Table S2 in the Supporting Information) for producing and making available their model output. For CMIP the U.S. Department of Energy's Program for Climate Model Diagnosis and Intercomparison provides coordinating support and led development of software infrastructure in partnership with the Global Organization for Earth System Science Portals. The source codes for the models used in this study are publicly available as follows: MetSim v2.0.0 — https://github.com/UW-Hydro/MetSim; VIC v.5 — https://github.com/UW-Hydro/VIC; MOSART-WM — https://github.com/IMMM-SFA/wm; RBM-2L — https://github.com/UW-Hydro/RBM/tree/RBM_res_develop.

Reference

Abatzoglou, J. T., & Brown, T. J. (2012). A comparison of statistical downscaling methods suited for wildfire applications. International Journal of Climatology, 32(5), 772-780. https://doi.org/10.1002/joc.2312

Abatzoglou, J.T. (2013). Development of gridded surface meteorological data for ecological applications and modelling. International Journal of Climatology, 33(1), 121-131. https://doi.org/10.1002/joc.3413

Arora, R., Toffolon, M., Tockner, K., & Venohr, M. (2018). Thermal discontinuities along a lowland river: The importance of urban areas and lakes. Journal of hydrology, 564, 811-823. https://doi.org/10.1016/j.jhydrol.2018.05.066

Averyt, K., Macknick, J., Rogers, J., Madden, N., Fisher, J., Meldrum, J., & Newmark, R. (2013). Water use for electricity in the United States: an analysis of reported and calculated water use information for 2008. Environmental Research Letters, 8(1), 015001. http://dx.doi.org/10.1088/1748-9326/8/1/015001

Bartos, M. D., & Chester, M. V. (2015). Impacts of climate change on electric power supply in the Western United States. Nature Climate Change, 5(8), 748-752. https://doi.org/10.1038/nclimate2648

Bennett, A., Hamman, J., & Nijssen, B. (2020). MetSim: A Python package for estimation and disaggregation of meteorological data. Journal of Open Source Software, 5(47), 2042. https://doi.org/10.21105/joss.02042

Boehlert, B., Strzepek, K. M., Chapra, S. C., Fant, C., Gebretsadik, Y., Lickley, M., et al. (2015). Climate change impacts and greenhouse gas mitigation effects on U.S. water quality.

675 Journal of Advances in Modeling Earth Systems, 7, 1326–1338.
676 https://doi.org/10.1002/2014MS000400
677 Bohn, T. J., Livneh, B., Oyler, J. W., Running, S. W., Nijssen, B., & Lettenmaier, D. P. (2013).
678 Global evaluation of MTCLIM and related algorithms for forcing of ecological and
679 hydrological models. *Agricultural and Forest Meteorology*, *176*, 38–49.
680 https://doi.org/10.1016/j.agrformet.2013.03.003
681 Cai, H., Piccolroaz, S., Huang, J., Liu, Z., Liu, F., & Toffolon, M. (2018). Quantifying the
682 impact of the Three Gorges Dam on the thermal dynamics of the Yangtze River.
683 Environmental Research Letters, 13(5), 054016. https://doi.org/10.1088/1748-
684 9326/aab9e0
685 Chapra, S. C. (1997). Surface water-quality modeling. Long Grove, IL: Waveland Press.
686 Cole, T. M., & Wells, S. A. (2015). CE-QUAL-W2: A two-dimensional, laterally averaged,
687 hydrodynamic and water quality model, Version 4.0. Portland State University, Portland,
688 OR, USA.
689 Devore, J.L. (2011). Probability and Statistics for Engineering and the Sciences. Boston, MA:
690 Cengage learning.
691 Gelda, R. K., Owens, E. M., & Effler, S. W. (1998). Calibration, verification, and an application
692 of a two-dimensional hydrothermal model [CE-QUAL-W2 (t)] for Cannonsville
693 Reservoir. Lake and Reservoir Management, 14(2-3), 186-196.
694 https://doi.org/10.1080/07438149809354330
695 Hamman, J. J., Nijssen, B., Bohn, T.J., Gergel, D. R., & Mao, Y. (2018). The Variable
696 Infiltration Capacity model version 5 (VIC-5): infrastructure improvements for new

applications and reproducibility. Geoscientific Model Development, 11(8), 3481-3496. https://doi.org/10.5194/gmd-11-3481-2018

Hanna, R. B., Saito, L., Bartholow, J. M., & Sandelin, J. (1999). Results of simulated temperature control device operations on in-reservoir and discharge water temperatures using CE-QUAL-W2. Lake and Reservoir Management, 15(2), 87-102. https://doi.org/10.1080/10.1080/07438149909353954

Hejazi, M. I., Voisin, N., Liu, L., Bramer, L. M., Fortin, D. C., Hathaway, J. E., Huang, M., Kyle, P., Leung, L. R., Li, H., & Liu, Y. (2015). 21st century United States emissions mitigation could increase water stress more than the climate change it is mitigating. Proceedings of the National Academy of Sciences, 112(34), 10635-10640. https://doi.org/10.1073/pnas.1421675112

Isaak, D. J., Wollrab, S., Horan, D., & Chandler, G. (2012). Climate change effects on stream and river temperatures across the northwest US from 1980–2009 and implications for salmonid fishes. Climatic Change, 113(2), 499-524. https://doi.org/10.1007/s10584-011-0326-z

Kędra, M., & Wiejaczka, Ł. (2018). Climatic and dam-induced impacts on river water temperature: assessment and management implications. Science of the Total Environment, 626, 1474-1483. https://doi.org/10.1016/j.scitotenv.2017.10.044

Lehner, B., Liermann, C. R., Revenga, C., Vörösmarty, C., Fekete, B., Crouzet, P., Döll, P., Endejan, M., Frenken, K., Magome, J., & Nilsson, C. (2011). High-resolution mapping of the world's reservoirs and dams for sustainable river-flow management. Frontiers in Ecology and the Environment, 9(9), 494-502. https://doi.org/10.1890/100125

719 Li, H., Wigmosta, M. S., Wu, H., Huang, M., Ke, Y., Coleman, A. M., & Leung, L. R. (2013). A
720 physically based runoff routing model for land surface and earth system models. Journal
721 of Hydrometeorology, 14(3), 808-828. https://doi.org/10.1175/JHM-D-12-015.1
722 Li, H., Leung, L. R., Tesfa, T., Voisin, N., Hejazi, M., Liu, L., Liu, Y., Rice, J., Wu, H. & Yang,
723 X. (2015). Modeling stream temperature in the Anthropocene: An earth system modeling
724 approach. Journal of Advances in Modeling Earth Systems, 7(4), 1661-1679.
725 https://doi.org/10.1002/2015MS000471
726 Liang, X., Lettenmaier, D. P., Wood, E. F. and Burges, S. J., 1994. A simple hydrologically
727 based model of land surface water and energy fluxes for general circulation models.
728 Journal of Geophysical Research: Atmospheres, 99(D7), 14415-14428.
729 https://doi.org/10.1029/94JD00483
730 Liu, L., Hejazi, M., Li, H., Forman, B., & Zhang, X. (2017). Vulnerability of US thermoelectric
731 power generation to climate change when incorporating state-level environmental
732 regulations. Nature Energy, 2(8), 17109. https://doi.org/10.1038/nenergy.2017.109
733 Lowney, C. L. (2000). Stream temperature variation in regulated rivers: Evidence for a spatial
734 pattern in daily minimum and maximum magnitudes. Water Resources Research, 36(10),
735 2947-2955. https://doi.org/10.1029/2000WR900142
736 Maheu, A., St-Hilaire, A., Caissie, D., & El-Jabi, N. (2016). Understanding the thermal regime
737 of rivers influenced by small and medium size dams in Eastern Canada. River research
738 and applications, 32(10), 2032-2044. https://doi.org/10.1002/rra.3046
739 Mantua, N., Tohver, I. & Hamlet, A. (2010). Climate change impacts on streamflow extremes
740 and summertime stream temperature and their possible consequences for freshwater

741 salmon habitat in Washington State. Climatic Change, 102(1-2), 187-223.
742 https://doi.org/10.1007/s10584-010-9845-2
743 Maurer, E. P., Wood, A. W., Adam, J. C., Lettenmaier, D. P., & Nijssen, B. (2002). A long-term
744 hydrologically based dataset of land surface fluxes and states for the conterminous United
745 States. Journal of climate, 15(22), 3237-3251. https://doi.org/10.1175/1520-
746 0442(2002)015<3237:ALTHBD>2.0.CO;2
747 McCall, J., Macknick, J., & Macknick, J. (2016). Water-related power plant curtailments: An
748 overview of incidents and contributing factors (No. NREL/TP-6A20-67084). National
749 Renewable Energy Lab.(NREL), Golden, CO (United States). Retrieved from:
750 https://www.osti.gov/biblio/1338176
751 Miara, A., Macknick, J. E., Vörösmarty, C. J., Tidwell, V. C., Newmark, R., & Fekete, B.
752 (2017). Climate and water resource change impacts and adaptation potential for US
753 power supply. Nature Climate Change, 7(11), 793. https://doi.org/10.1038/nclimate3417
754 Miara A., Vörösmarty, C. J., Macknick, J. E., Tidwell, V.C., Fekete B., Corsi F., & Newmark, R.
755 (2018). Thermal pollution impacts on rivers and power supply in the Mississippi River
756 watershed. Environmental Research Letters. 13(3):034033. https://doi.org/10.1088/1748-
757 9326/aaac85
758 Mohseni, O., Stefan, H.G., & Erickson, T.R. (1998). A nonlinear regression model for weekly
759 stream temperatures. Water Resources Research, 34(10), 2685-2692.
760 https://doi.org/10.1029/98WR01877
761 Niemeyer, R. J., Cheng, Y., Mao, Y., Yearsley, J. R. & Nijssen, B. (2018). A Thermally
762 Stratified Reservoir Module for Large-Scale Distributed Stream Temperature Models

With Application in the Tennessee River Basin. Water Resources Research, 54(10), 8103-8119. https://doi.org/10.1029/2018WR022615

Raptis, C. E., van Vliet, M. T., & Pfister, S. (2016). Global thermal pollution of rivers from thermoelectric power plants. Environmental Research Letters, 11(10), 104011. https://doi.org/10.1088/1748-9326/11/10/104011

Rheinheimer, D. E., Null, S. E. & Lund, J. R. (2014). Optimizing selective withdrawal from reservoirs to manage downstream temperatures with climate warming. Journal of Water Resources Planning and Management, 141(4), 04014063. https://doi.org/10.1061/(ASCE)WR.1943-5452.0000447

Strzepek, K., Fant, C., Gebretsadik, Y., Lickley, M., Boehlert, B., Chapra, S., ... & Schlosser, C. A. (2015). Water body temperature model for assessing climate change impacts on thermal cooling (No. 280). MIT Joint Program on the Science and Policy of Global Change, Cambridge, MA (United States). Retrieved from: https://dspace.mit.edu/handle/1721.1/99410

Sun, N., Yearsley, J., Voisin, N., & Lettenmaier, D. P. (2015). A spatially distributed model for the assessment of land use impacts on stream temperature in small urban watersheds. Hydrological Processes, 29(10), 2331-2345. https://doi.org/10.1002/hyp.10363

Taylor, K. E., Stouffer, R. J. & Meehl, G. A. (2011). An Overview of CMIP5 and the Experiment Design. Bulletin of the American Meteorological Society, 93, https://doi.org/10.1175/bams-d-11-00094.1.

Thornton, P. E., & Running, S. W. (1999). An improved algorithm for estimating incident daily solar radiation from measurements of temperature, humidity, and precipitation.

Agricultural and Forest Meteorology, *93*(4), 211–228. https://doi.org/10.1016/S0168-1923(98)00126-9

United States Geological Survey (USGS) (2019). National Water Information System data available on the World Wide Web (USGS Water Data for the Nation) [available at https://waterdata.usgs.gov/nwis/]

USACE-HEC (U. S. Army Corps of Engineers-Hydrologic Engineering Center) (1986). WQRRS Water Quality for River-Reservoir Systems user's manual, Hydrologic Engineering Center, Davis, California.

van Vliet, M. T. H., Yearsley, J. R., Ludwig, F., Vögele, S., Lettenmaier, D. P., & Kabat, P. (2012). Vulnerability of US and European electricity supply to climate change. Nature Climate Change, 2(9), 676–681. https://doi.org/10.1038/nclimate1546

van Vliet, M. T. H., Vögele, S., & Rübbelke. D. (2013) Water constraints on European power supply under climate change: impacts on electricity prices. Environmental Research Letters, 8(3), 035010. https://doi.org/10.1088/1748-9326/8/3/035010

van Vliet, M. T. H., Wiberg, D., Leduc, S., & Riahi, K. (2016). Power-generation system vulnerability and adaptation to changes in climate and water resources. Nature Climate Change, 6(4), 375-380. https://doi.org/10.1038/nclimate2903

Voisin, N., Li, H., Ward, D., Huang, M., Wigmosta, M. & Leung, L. R. (2013a). On an improved sub-regional water resources management representation for integration into earth system models. Hydrology and Earth System Sciences, 17(9), 3605-3622. https://doi.org/10.5194/hess-17-3605-2013

Voisin, N., Liu, L., Hejazi, M., Tesfa, T., Li, H., Huang, M., Liu, Y., & Leung, R. L. (2013b) One-way coupling of an integrated assessment model and a water resources model:

evaluation and implications of future changes over the US Midwest, Hydrology and Earth System Sciences 17(11), 4555-4575. https://doi.org/10.5194/hess-17-4555-2013

Voisin, N., Leung, R., & Hejazi, M. (2016) Drivers of change in managed water resources: Modeling the impacts of climate and socio-economic changes using the U.S. Midwest as a case study. Terrestrial Water Cycle and Climate Change: Natural and Human-Induced Impacts, 221, 169.

Voisin, N., Hejazi, M. I., Leung, L. R., Liu, L., Huang, M., Li, H. Y., & Tesfa, T. (2017). Effects of spatially distributed sectoral water management on the redistribution of water resources in an integrated water model. Water Resources Research, 53(5), 4253-4270. https://doi.org/10.1002/2016WR019767

Willey, R. G. (1986). HEC-5Q: System water quality modeling. Davis, Calif.: U.S. Army Corps of Engineers, Hydrologic Engineering Center.

Yearsley, J. R. (2009). A semi-Lagrangian water temperature model for advection-dominated river systems. Water Resources Research, 45, W12405. https://doi.org/10.1029/2008WR007629

Yearsley, J. R. (2012). A grid-based approach for simulating stream temperature. Water Resources Research, 48, W03506. https://doi.org/10.1029/2011WR011515

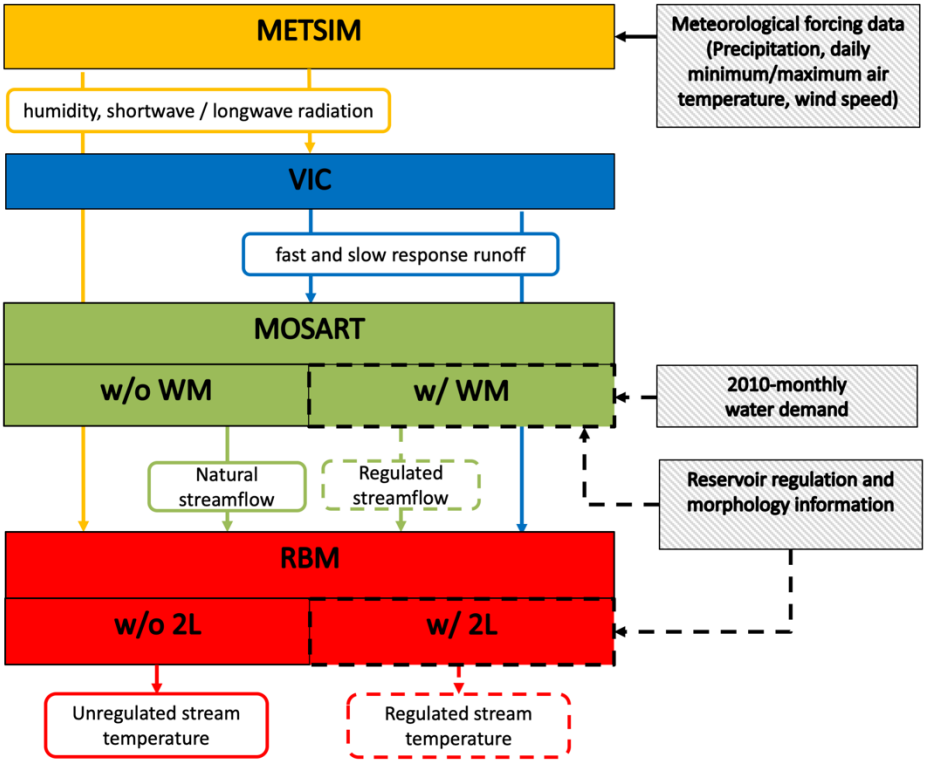
Yearsley, J.Sun, N., Baptiste, M., & Nijssen, B. (2019). Assessing the impacts of hydrologic and land use alterations on water temperature in the Farmington River basin in Connecticut. Hydrology and Earth System Sciences, 23(11), 4491-4508. https://doi.org/10.5194/hess-23-4491-2019

Yigzaw, W., Li, H. Y., Demissie, Y., Hejazi, M. I., Leung, L. R., Voisin, N., & Payn, R. (2018). A New Global Storage-Area-Depth Data Set for Modeling Reservoirs in Land Surface

and Earth System Models. Water Resources Research, 54(12), 10-372. https://doi.org/10.1029/2017WR022040

Zhang, X., Li, H.-Y., Leung, L. R., Liu, L., Hejazi, M. I., Forman, B. A., & Yigzaw, W. (2020). River regulation alleviates the impacts of climate change on U.S. thermoelectricity production. Journal of Geophysical Research: Atmospheres, 125, e2019JD031618. https://doi.org/10.1029/2019JD031618

Zhou, T., Voisin, N., Leng, G., Huang, M., & Kraucunas, I. (2018). Sensitivity of Regulated Flow Regimes to Climate Change in the Western United States. Journal of Hydrometeorology, 19(3), 499-515. https://doi.org/10.1175/JHM-D-17-0095.1

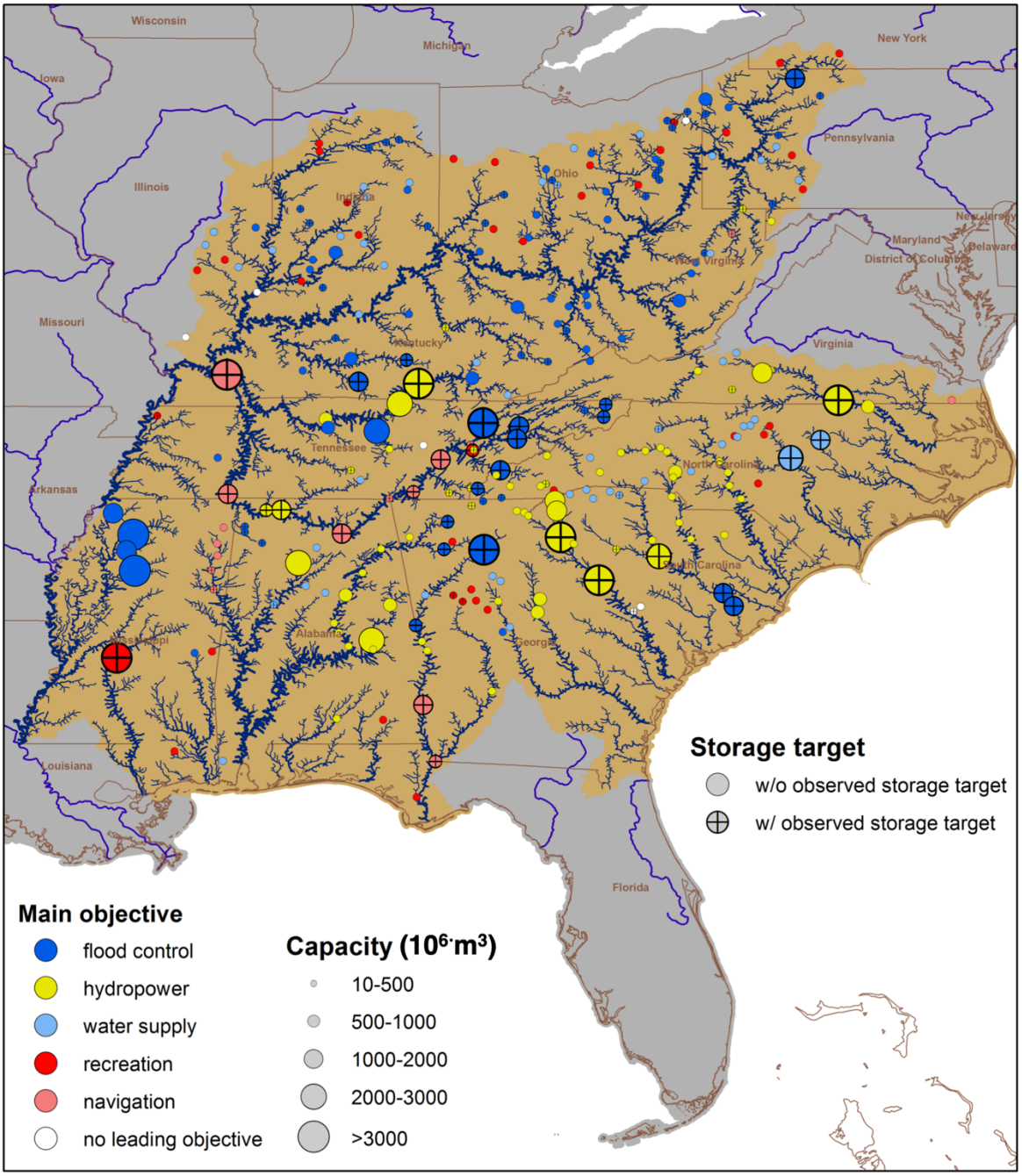


841

842 Figure 1: Model framework. Dashed boxes and arrows denote models and inputs that are only

843 used in the regulated model setup.

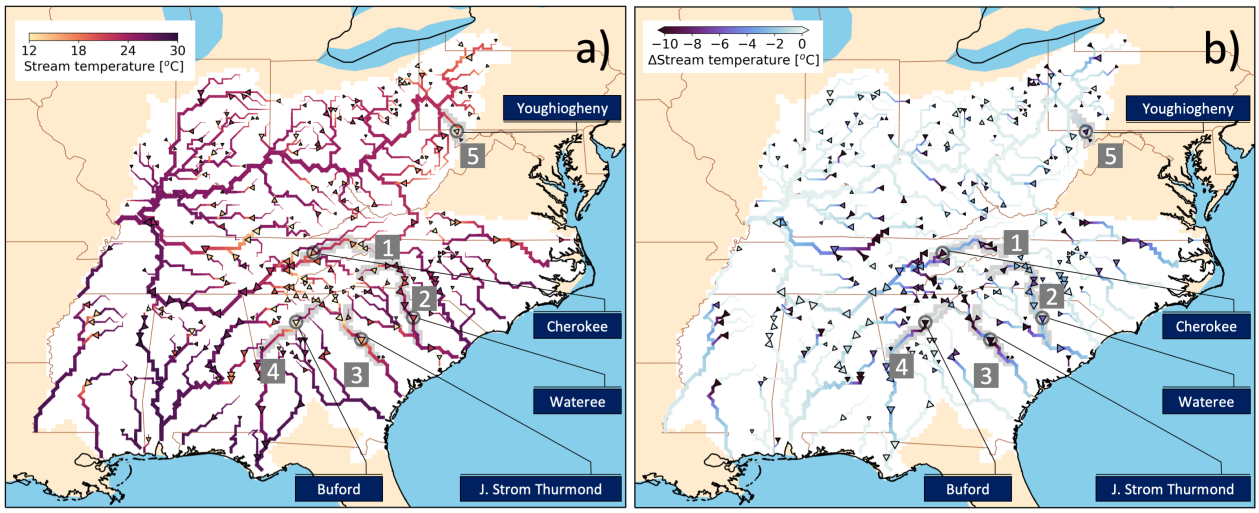
844



845

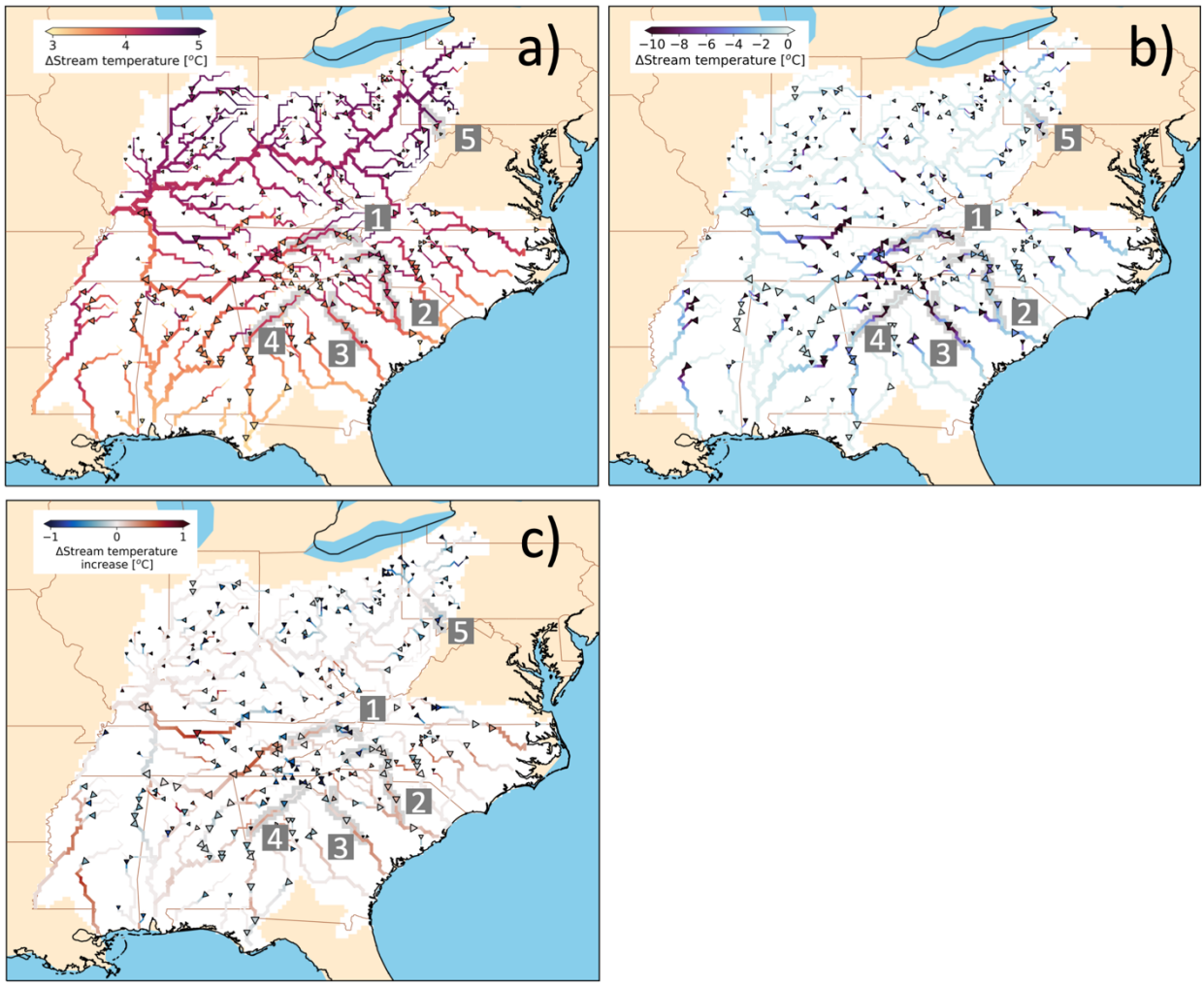
846 Figure 2: Study region. All reservoirs included in this study are represented by the circles.

847



848

849 Figure 3: (a) Historical mean summer river temperature for the regulated scenario; (b) difference
850 in historical mean summer temperature between the regulated and unregulated model setups.
851 Triangles denote reservoir locations and point downstream; gray-shaded river channels were
852 selected to show the temperature profile along the river, i.e., 1-Tennessee River, 2-Catawba
853 River, 3-Savannah River, 4-Chattahoochee River, 5-Youghiogheny River (see Figure 7).
854 Selected reservoirs discussed in the manuscript are circled with the dam names as shown.



855

856 Figure 4: Projected mean summer river temperature (a) increases in regulated river system

857 ($\Delta T_{reg,2080s-reg,hist}$), (b) differences between regulated and unregulated model setups in 2080s

858 ($\Delta T_{reg,2080s-nat,2080s}$), and (c) projected changes in the median reservoir impacts on

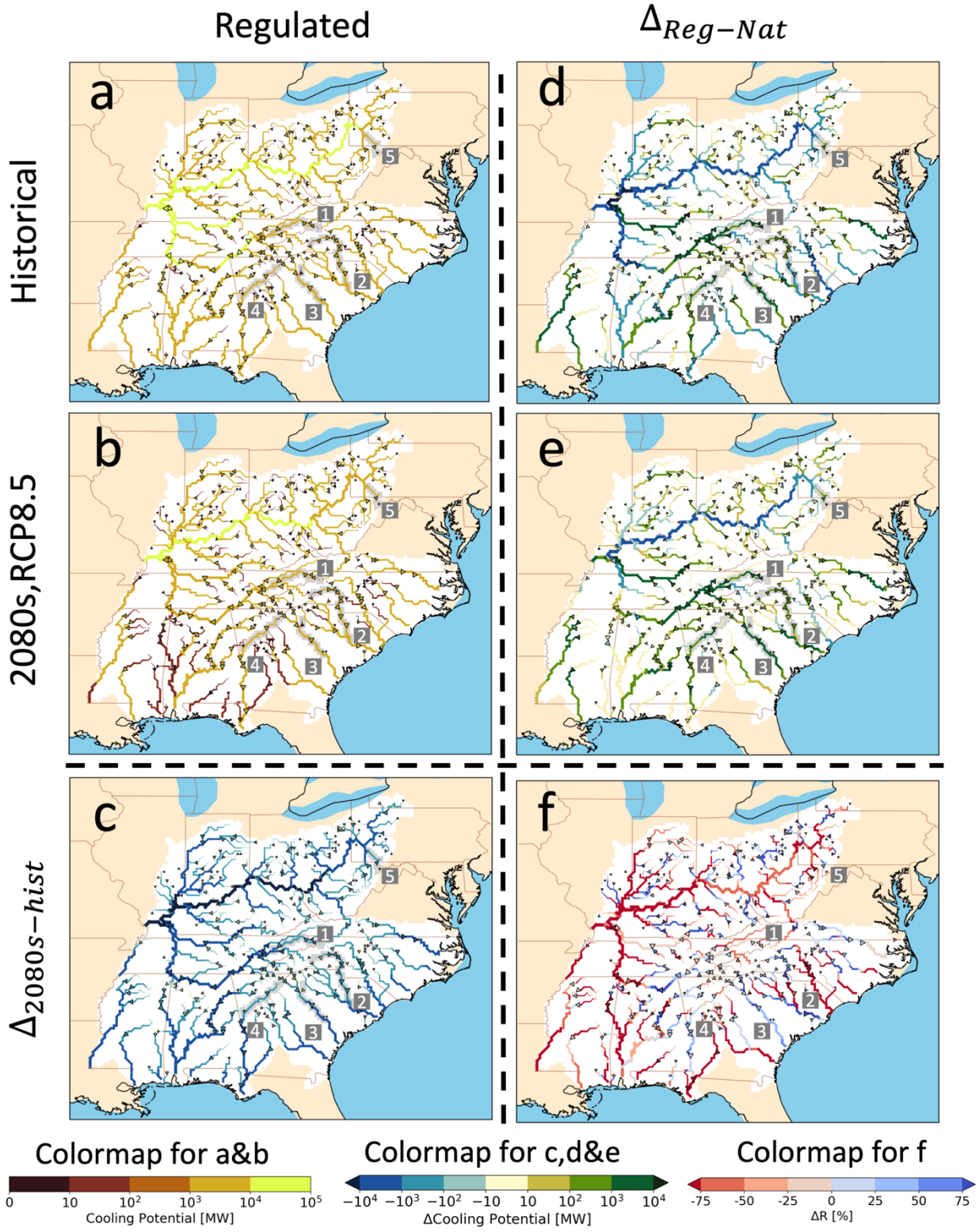
859 temperatures between the 2080s under RCP8.5 and historical period, i.e.,

860 $\Delta(\Delta T_{reg,2080s-nat,2080s} - \Delta T_{reg,hist-nat,hist})$. We highlight the same five rivers as in Figure 3:

861 1-Tennessee River, 2-Catawba River, 3-Savannah River, 4-Chattahoochee River, 5-

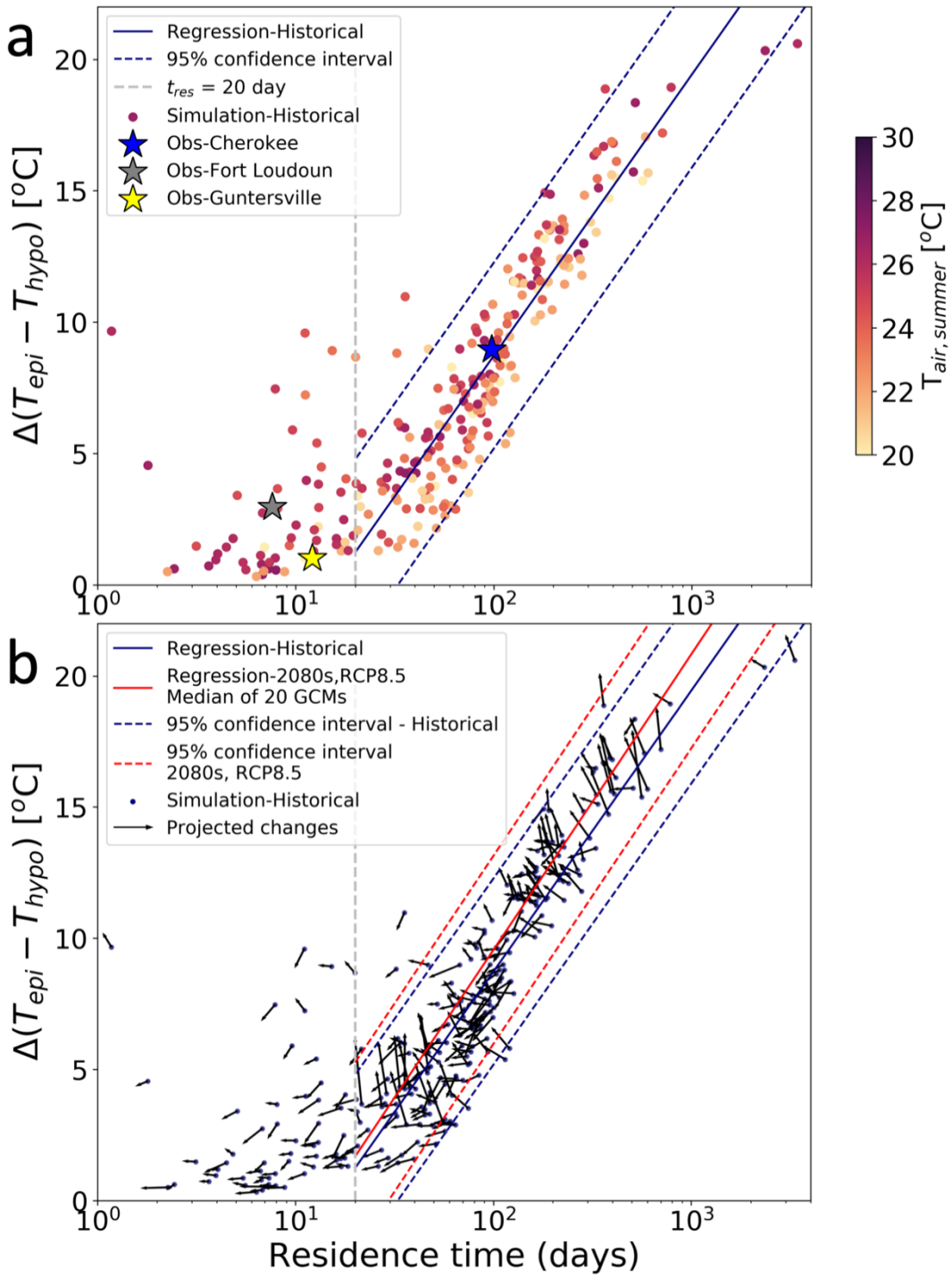
862 Youghiogheny River.

863


Regulated
$\Delta_{Reg-Nat}$
Historical
2080s,RCP8.5
$\Delta_{2080s-hist}$
a
b
c
d
e
f
1
2
3
4
5
Colormap for a&b
0 10 10^2 10^3 10^4 10^5
Cooling Potential [MW]
Colormap for c,d&e
-10^4 -10^3 -10^2 -10 10 10^2 10^3 10^4
ΔCooling Potential [MW]
Colormap for f
-75 -50 -25 0 25 50 75
ΔR [%]

864

865 Figure 5: Spatial maps of regulated cooling potential for (a) historical period and (b) 2080s under
866 RCP8.5; climate change impacts on regulated cooling potential under RCP8.5 (c,
867 $\Delta E_{cp(reg,2080s-reg,hist)}$); difference of cooling potential between regulated and unregulated
868 model setups for (d) historical period and (e) 2080s under RCP8.5 and (f) the change in the effect
869 of regulation on cooling between the future and historic periods (Eq. 2).



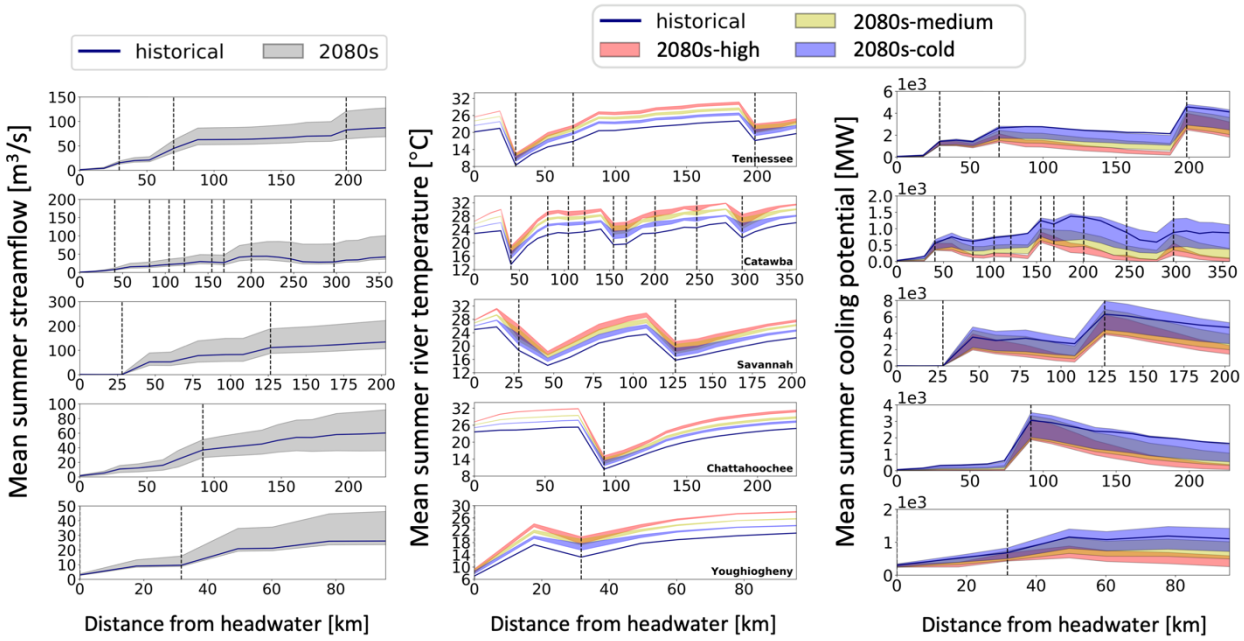
870

871 Figure 6: (a) Simulated reservoir thermal stratification versus residence time during the historical

872 period. Each dot represents the simulated result for one reservoir during the historical period.

873 The dot color represents the mean air temperature at each reservoir location. Stars represent the

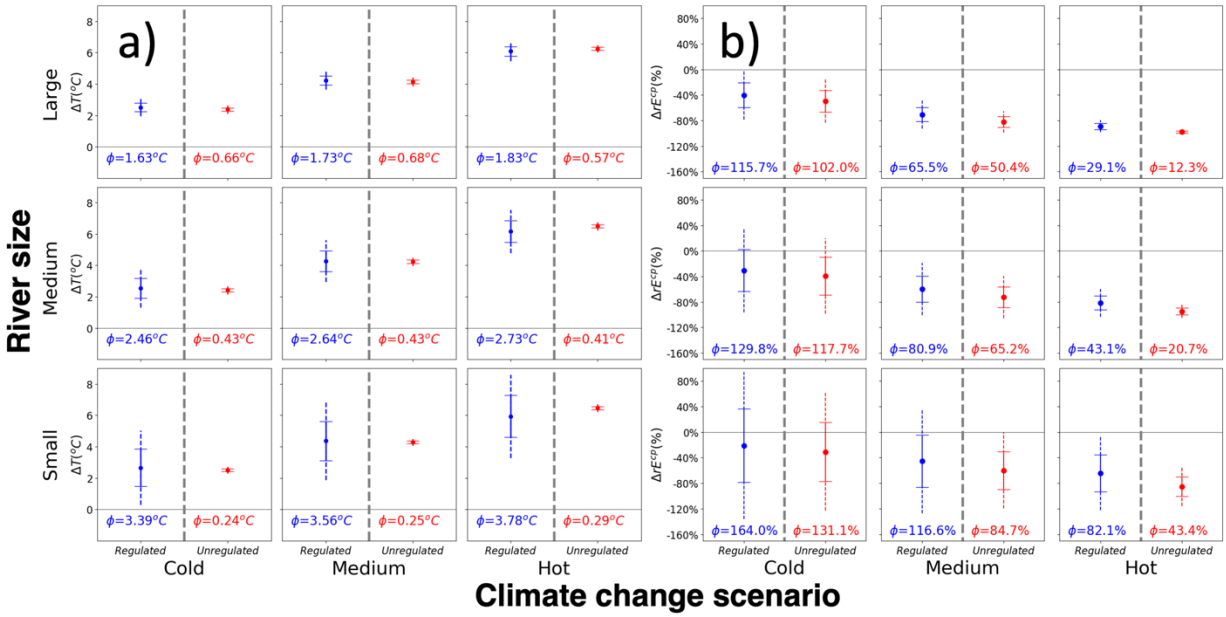
874 observations, with blue, grey and yellow stars representing Cherokee, Fort Loudoun and
875 Guntersville Reservoirs, respectively. (b) Climate change impacts on the relationship between
876 thermal stratification and residence time. Each dot is the same as in (a) with an arrow pointing to
877 the median simulated result (20 GCMs) by the 2080s under RCP8.5 for the same reservoir.
878



879

880 **Figure 7**: Sensitivity experiment: mean summer streamflow (left), stream temperature (middle),
881 and cooling potential (right) profiles for the historical period and the 2080s under RCP8.5 along
882 five selected river channels, i.e., Tennessee River, Catawba River, Savannah River,
883 Chattahoochee River, and Youghiogheny River (from top to bottom). In the left panel, the grey
884 shaded area denotes ranges of projected mean summer streamflow. In the middle and right
885 panels, the shaded areas denote the ranges in projected mean summer river temperature and
886 cooling potential, respectively. The vertical dashed lines denote dam locations.

887



888

889 **Figure 8:** Projected changes in (a) mean summer river temperature and (b) mean summer

890 cooling potential by river size and climate changes scenarios. Estimates are provided for river

891 segments subject to regulation (*m*=true) for both the regulated (*r*=reg; blue) and unregulated

892 (*r*=unreg; red) model setups. The solid and dashed bands around the mean denote one and two

893 times the error in the estimates ($\hat{\sigma}$ in Eqn. 6) as a result of hydrological errors.